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Expt.No: 4

Title: Python program to perform Basic operations on an Image

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In [1]: 1 import cv2
2 import matplotlib.pyplot as plt
3
4 image = cv2.imread('ResizeImage3.jpg')
5
6 x, y, w, h = 100, 100, 300, 300
7 cropped_image = image[y:y+h, x:x+w]
8
9 scaled_image = cv2.resize(image, (800, 800))
10
11 rotated_image = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)
12
13 flipped_image_horizontal = cv2.flip(image, 1)
14 flipped_image_vertical = cv2.flip(image, 0)
15
16 plt.figure(figsize=(12, 8))
17
18 plt.subplot(2, 3, 1)
19 plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
20 plt.title('Original Image')
21 plt.axis('off')
22
23 plt.subplot(2, 3, 2)
24 plt.imshow(cv2.cvtColor(cropped_image, cv2.COLOR_BGR2RGB))
25 plt.title('Cropped Image')
26 plt.axis('off')
27
28 plt.subplot(2, 3, 3)
29 plt.imshow(cv2.cvtColor(scaled_image, cv2.COLOR_BGR2RGB))
30 plt.title('Scaled Image')
31 plt.axis('off')
32
33 plt.subplot(2, 3, 4)
34 plt.imshow(cv2.cvtColor(rotated_image, cv2.COLOR_BGR2RGB))
35 plt.title('Rotated Image')
36 plt.axis('off')
37
38 plt.subplot(2, 3, 5)
39 plt.imshow(cv2.cvtColor(flipped_image_horizontal, cv2.COLOR_BGR2RGB))
40 plt.title('Flipped Image (Horizontal)')
41 plt.axis('off')
42
43 plt.subplot(2, 3, 6)
44 plt.imshow(cv2.cvtColor(flipped_image_vertical, cv2.COLOR_BGR2RGB))
45 plt.title('Flipped Image (Vertical)')
46 plt.axis('off')
47 plt.show()
```

Original Image



Cropped Image



Scaled Image



Rotated Image



Flipped Image (Horizontal)



Flipped Image (Vertical)



In []:

1